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slot-B

1.) Mathematical Formulation of formal language and Automata.

1.) Input Alphabet

The input alphabet is the set of symbols used to create access codes

$\Sigma = \{0, 1\}$ Here 0 & 1 represent binary symbol.

2.) Possible string and formal language

A string is combination of symbols from the alphabet

A formal language is set of strings that statisfy a particular condition

Eg:- $L = \{w \in \{0,1\}^* \mid w \text{ contain } 101\}$

3) Difference between strings and languages

* Finite string: A limited sequence of symbols
* Empty string: A string containing no symbols It is represented by ε
* Language: A collection of string.

4) finite-Tuple Representation.

A finite Automaton is mathematically represented using five elements $M = (Q, \Sigma, \delta, q_0, F)$

conclusion:- Formal language and finite automata help the university validate binary access codes accurately and seamlessly.

2) Task 2 :- Mathematical Design and Construction of DFA

Aim :- To Construct a Deterministic finite Automata that accepts binary string started and ending with same symbol.

1) Required states

* q_s - initial state * q_0 - starts with 0 ends with 0
* q_1 - starts with 1 & ends with 1
* q_{01} - starts with 0 & ends with 1
* q_{10} - starts with 1 & ends with 0

2) Five Tuple Representation

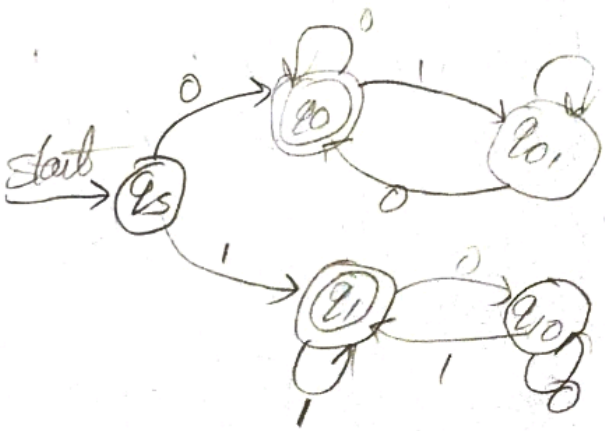
$M = (Q, \Sigma, \delta, q, S, F)$ $Q = \{q_s, q_0, q_1, q_{11}, q_{10}\}$

$F = \{q_0, q_1\}$

3) Transition Table :-

State	i/p	o/p
q_s	q_0	q_1
q_0	q_0	q_{01}
q_1	q_{10}	q_1
q_{01}	q_0	q_{01}
q_{10}	q_{10}	q_1

4) Diagram.



5) Sample input Traces

1) Input 00 : $q_s \rightarrow q_0 \rightarrow q_0$ = Accepted
2) Input 11 : $q_s \rightarrow q_1 \rightarrow q_1$ = Accepted
3) Input 01 : $q_s \rightarrow q_0 \rightarrow q_{01}$ = Rejected
4) Input 10 : $q_s \rightarrow q_1 \rightarrow q_{10}$ = Rejected

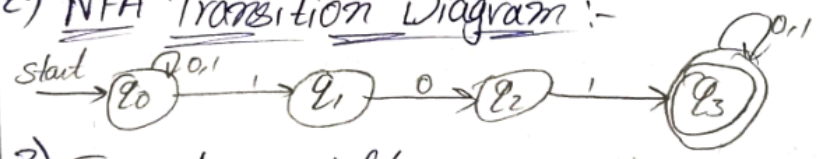
Task :- NON-Deterministic Finite Automata for Pattern Recognition.

Aim :- To design an NFA that accepts binary ac... of 0 code containing the substring 101 ...01, 101, 1101

1) Required states :-

* q_0 - Initial state; searches for the Pattern
* q_1 - Recogniges the first symbol1 ...t least
* q_2 - Recogniges the Pattern 10 ...2 of 3
* q_3 - Recogniges the complete Pattern 101 ...3 at

2) NFA Transition Diagram :-



3) Transition table :-

state	i/p(i)	i/p
q_0	i/o(i)	i/p(1)
q_1		$\emptyset$
q_2	$\emptyset$	
q_3		

4) Input Processing :-

1) Input 101 : $q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$ = Accepted.
2) Input 1101 : $q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$ = Accepted
3) Input 01010 : Accepted because it contains 101
4) Input 0000 : Rejected because it does not contain...

4) Task4- Equivalence of NFA and DFA using subset Construction

Aim:- To convert the NFA from Tasks 3 into an equivalent DFA using the subsets construction method.

1.) NFA states:- $Q = \{q_0, q_1, q_2, q_3\}$

2.) Possible DFA states

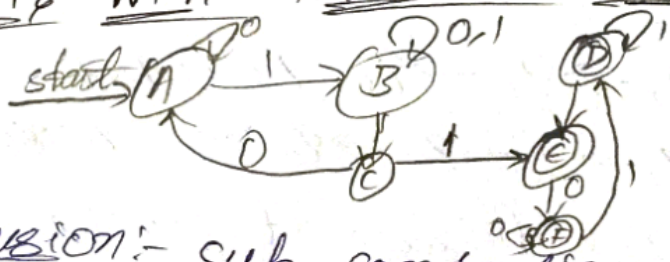
$A = \{q_0\}$ $B = \{q_0, q_1\}$ $C = \{q_0, q_2\}$ $D = \{q_0, q_1, q_3\}$

$E = \{q_0, q_2, q_3\}$ $F = \{q_3\}$

3.) subset construction Transition Table:-

DFA states	i/p(0)	i/p(1)
→ A	A	B
B	C	B
C	A	D
D	E	D
E	F	D
F	F	D

4.) DFA & NFA Transition Diagram



Conclusion:- sub construction converts an NFA int an equivalent DFA. Both automata accept

A Regular Expression is used to represent ... time it helps identify whether

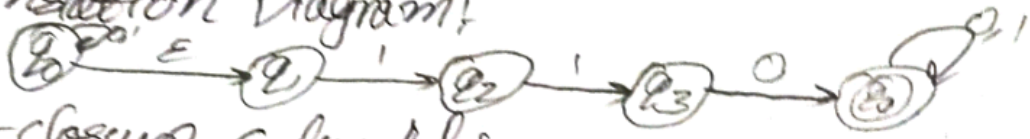
Task 5:- ε-NFA Construction, ε-closure and Equivalence Analysis

Aim:- To Construct an ε-NFA that accepts binary access codes containing the Pattern 110 & convert it into an equivalent-NFA.

1) Required states:-

* q_0 - initial states
* q_1 - searches for the first 1
* q_2 - Recognises 11
* q_3 - Recognises 110

2) Transition Diagram:-



3) ε-closure calculation

ε-closure means the set of states reachable Zero or more ε-transitions.

$\varepsilon\text{-closure}(q_0) = \{q_0, q_1\}$ $\varepsilon\text{-closure}(q_1) = \{q_1\}$

$\varepsilon\text{-closure}(q_2) = \{q_2\}$ $\varepsilon\text{-closure}(q_3) = \{q_3\}$

$\varepsilon\text{-closure}(q_4) = \{q_4\}$

4) Transition table:-

State	i/p(0)	i/p(1)	ε
q_0			
q_1	∅		
q_2	∅		∅
q_3		∅	∅
q_4		∅	∅

5) Equivalent NFA without ε-Trans

state	i/p(0)	i/p(1)
q_0		
q_1	∅	
q_2	∅	
q_3		∅
q_4		

6) Sample input verification:-

1.) Input 110: Accepted
2.) input 1110 : Accepted
3.) input 01100 : Accepted
4.) input 010 : Rejected
5.) input 0000 : Rejected.

Conclusion:- ε-NFA can be converted into an equivalent NFA by calculating ε-closure and removing ε-transition.

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